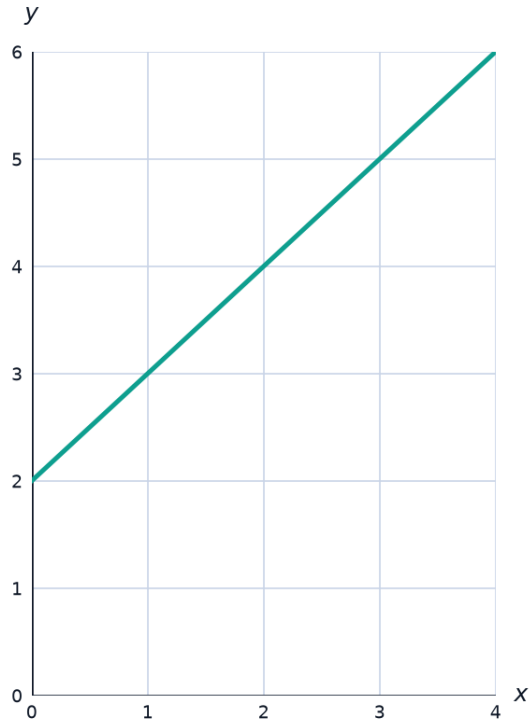


Pre-Calculus Concepts



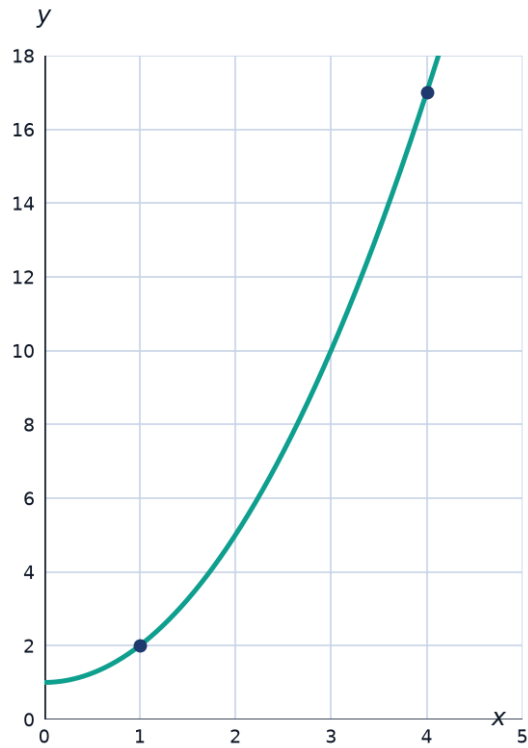
Limits: an introductory numerical approach

- 1 Estimate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression algebraically.



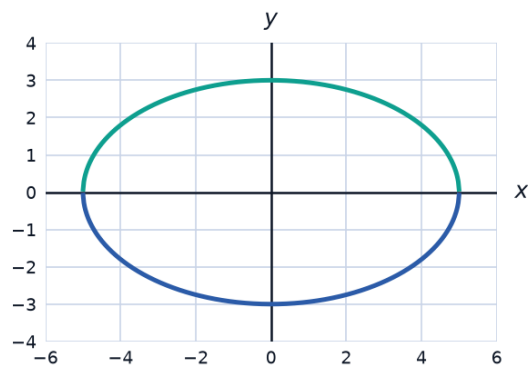
Average rate of change

- 2 Find the average rate of change of $f(x) = x^2 + 1$ over the interval $[1, 4]$.

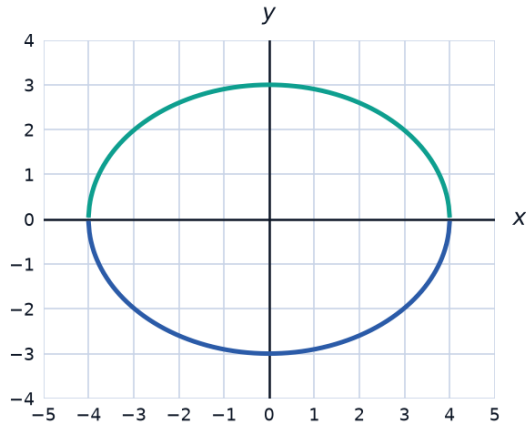


Conic sections: ellipses

- 3 Find the equation of an ellipse centered at the origin with $a = 5$ (horizontal semi-axis) and $b = 3$ (vertical semi-axis).

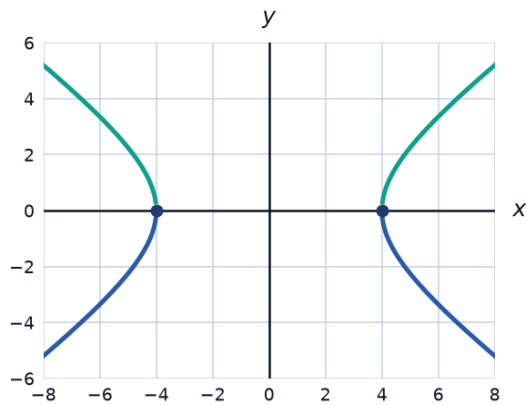


- 4 For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, find the coordinates of the vertices along the major axis.



Conic sections: hyperbolas

- 5 Find the equation of a hyperbola centered at the origin with vertices at $(\pm 4, 0)$ and $b = 3$.



Parametric equations

- 6 Eliminate the parameter from $x = t + 1$ and $y = t^2$, expressing y in terms of x .

Sequences and series: sigma notation

- 7 Evaluate $\sum_{k=1}^5 (2k + 1)$.

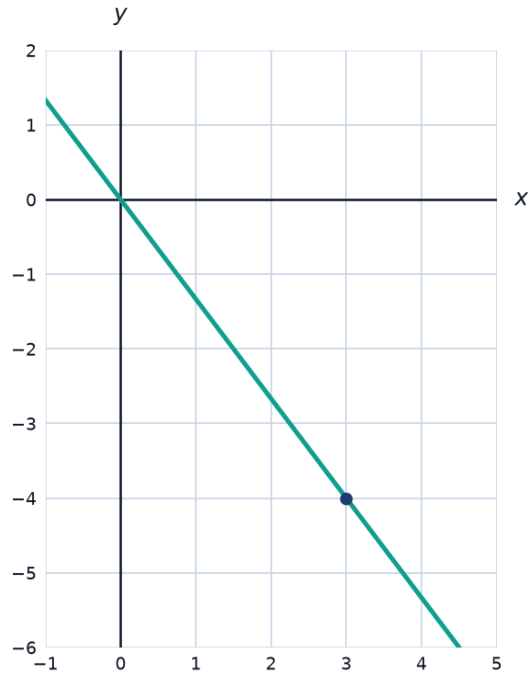
- 8 Write $4 + 8 + 12 + 16 + 20$ using sigma notation, and evaluate it.
-

Binomial theorem

- 9 Use the binomial theorem to expand $(x + 2)^4$.
- 10 Find the coefficient of x^3 in the expansion of $(x + 3)^5$, using $\binom{5}{k}$.
-

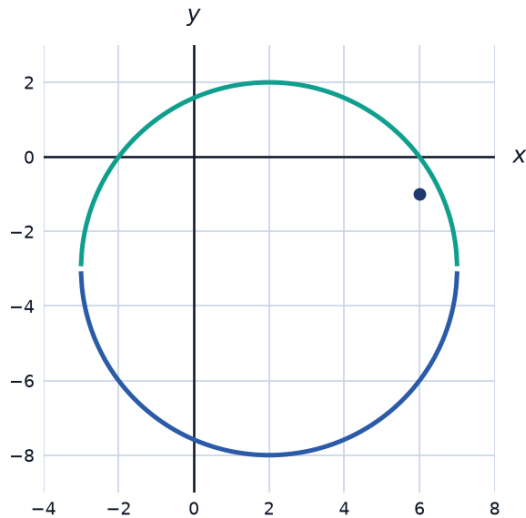
Vectors in the plane

- 11 Given $\mathbf{u} = (3, -4)$, find its magnitude and a unit vector in the same direction.



A multi-part conic and function analysis problem

- 12 This question has two parts. A circle has equation $x^2 + y^2 - 4x + 6y - 12 = 0$. (a) Complete the square to write the equation in standard form, and state the center and radius. (b) Determine whether the point $(6, -1)$ lies inside, on, or outside the circle.



Matrix operations

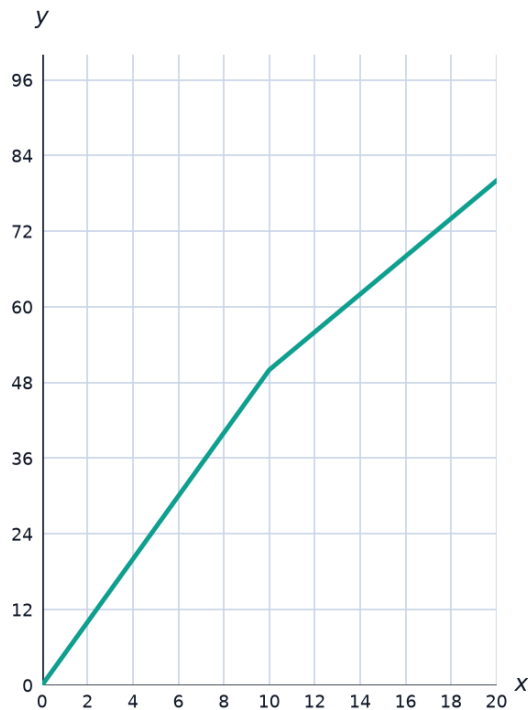
- 13 Matrix A has rows $(2, -1)$ and $(3, 4)$; matrix B has rows $(0, 5)$ and $(-2, 1)$. Find $A - B$.

Infinite geometric series

- 14 Find the sum of the infinite geometric series $8 + 4 + 2 + 1 + \dots$, using $S = \frac{a_1}{1 - r}$.
- 15 Write the repeating decimal $0.\overline{6}$ as a fraction, by treating it as an infinite geometric series $\frac{6}{10} + \frac{6}{100} + \frac{6}{1000} + \dots$.

Piecewise and step functions in applied contexts

- 16 A shipping company charges 5 dollars per pound for the first 10 pounds, then 3 dollars per pound for each pound beyond 10. Write a piecewise function $C(w)$ for the cost, and find the cost of a 15-pound package.

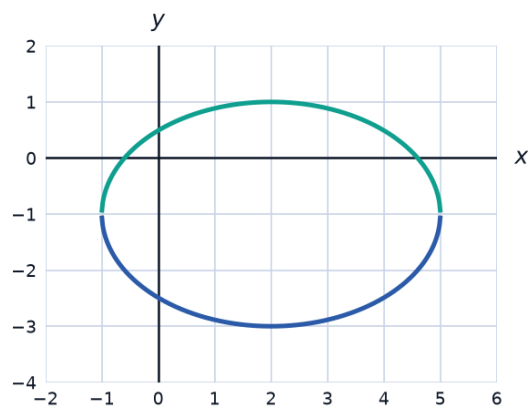


Finding terms of a sequence defined recursively

- 17 A sequence is defined by $a_1 = 3$ and $a_n = 2a_{n-1} + 1$ for $n \geq 2$. Find a_4 .

A multi-part conic section identification and analysis problem

- 18 This question has three parts. An equation is given: $4x^2 + 9y^2 - 16x + 18y - 11 = 0$. (a) Complete the square in both x and y to write the equation in standard conic form. (b) Identify the type of conic section. (c) State the center and the lengths of the semi-axes.



Multiple choice — easy

- 19 Find the average rate of change of $f(x) = 2x^2 - 3$ over the interval $[-3, 2]$.
- a. -10
 - b. 0
 - c. -2
 - d. -7
- 20 Find the average rate of change of $f(x) = 3x^2 - 7x + 5$ over the interval $[0, 3]$.
- a. 5
 - b. 2
 - c. 6
 - d. -1
- 21 Find the average rate of change of $f(x) = 2x^2 - 7x + 9$ over the interval $[1, 3]$.
- a. 3
 - b. -1
 - c. 2
 - d. 1
- 22 Find the average rate of change of $f(x) = 3x^2 - 7x - 9$ over the interval $[0, 4]$.
- a. 8
 - b. 1
 - c. 5
 - d. 20

- 23 Find the average rate of change of $f(x) = 3x^2 + 8x - 9$ over the interval $[5, 6]$.
- a. 41
 - b. 43
 - c. 40
 - d. 44
- 24 Find the average rate of change of $f(x) = 3x^2 - 5x - 1$ over the interval $[-2, -1]$.
- a. -15
 - b. -12
 - c. -11
 - d. -14
- 25 Find the average rate of change of $f(x) = x^2 + 7x$ over the interval $[-6, 3]$.
- a. 4
 - b. 5
 - c. 36
 - d. -5
- 26 Find the average rate of change of $f(x) = -x^2 - 3x - 3$ over the interval $[-4, 2]$.
- a. -7
 - b. -1
 - c. -2
 - d. -6

- 27 Find the average rate of change of $f(x) = x^2 - 9x + 7$ over the interval $[-3, 4]$.
- a. -8
 - b. -56
 - c. -15
 - d. -7
- 28 Find the average rate of change of $f(x) = -3x^2 + 7x - 2$ over the interval $[1, 4]$.
- a. -11
 - b. -6
 - c. -24
 - d. -8
- 29 Find the average rate of change of $f(x) = -3x^2 + 2x + 9$ over the interval $[0, 1]$.
- a. -4
 - b. -1
 - c. -2
 - d. 1
- 30 Find the average rate of change of $f(x) = -x^2 + 6x - 3$ over the interval $[1, 6]$.
- a. -1
 - b. -2
 - c. -5
 - d. -6

- 31 Find the average rate of change of $f(x) = x^2 - 7$ over the interval $[-2, 1]$.
- a. 0
 - b. -3
 - c. -4
 - d. -1
- 32 Find the average rate of change of $f(x) = -2x^2 - 6x + 7$ over the interval $[5, 6]$.
- a. -28
 - b. -29
 - c. -30
 - d. -26
- 33 Find the average rate of change of $f(x) = x^2 + 2x + 7$ over the interval $[3, 4]$.
- a. 11
 - b. 10
 - c. 8
 - d. 9

Multiple choice — medium

- 34 Evaluate $\sum_{k=1}^{10} (k - 1)$.
- a. 46
 - b. 44
 - c. 45
 - d. 55

35 Evaluate $\sum_{k=1}^4 (2k + 4)$.

a. 40

b. 34

c. 20

d. 36

36 Evaluate $\sum_{k=1}^{10} (2k - 2)$.

a. 90

b. 88

c. 100

d. 110

37 Evaluate $\sum_{k=1}^4 (k - 4)$.

a. -7

b. -2

c. 10

d. -6

38 Evaluate $\sum_{k=1}^{10} (k + 3)$.

a. 85

b. 84

c. 95

d. 55

39 Evaluate $\sum_{k=1}^{11} (k - 2)$.

a. 55

b. 66

c. 44

d. 43

40 Evaluate $\sum_{k=1}^8 (k - 5)$.

a. -5

b. 4

c. -4

d. 36

41 Evaluate $\sum_{k=1}^8 (k - 3)$.

a. 11

b. 12

c. 36

d. 20

42 Evaluate $\sum_{k=1}^{11} (2k + 1)$.

a. 132

b. 154

c. 143

d. 141

43 Evaluate $\sum_{k=1}^6 (-3k - 2)$.

a. -69

b. -72

c. -75

d. -63

44 Evaluate $\sum_{k=1}^{11} (-3k + 2)$.

a. -176

b. -173

c. -165

d. -198

45 Evaluate $\sum_{k=1}^8 (-2k + 5)$.

a. -24

b. -30

c. -72

d. -32

46 Evaluate $\sum_{k=1}^{10} (3k - 3)$.

a. 135

b. 145

c. 165

d. 132

47 Evaluate $\sum_{k=1}^{12} (3k + 5)$.

- a. 306
- b. 291
- c. 294
- d. 234

48 Evaluate $\sum_{k=1}^5 (-3k + 2)$.

- a. -32
- b. -45
- c. -35
- d. -30

Multiple choice — hard

49 For the ellipse $\frac{x^2}{1681} + \frac{y^2}{81} = 1$ (horizontal major axis), find the coordinates of the foci.

- a. $(\pm 9, 0)$
- b. $(0, \pm 40)$
- c. $(\pm 40, 0)$
- d. $(\pm 41, 0)$

50 For the ellipse $\frac{x^2}{841} + \frac{y^2}{400} = 1$ (horizontal major axis), find the coordinates of the foci.

- a. $(\pm 29, 0)$
- b. $(0, \pm 21)$
- c. $(\pm 21, 0)$
- d. $(\pm 20, 0)$

- 51 For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ (horizontal major axis), find the coordinates of the foci.
- a. $(\pm 4, 0)$
 - b. $(\pm 3, 0)$
 - c. $(0, \pm 4)$
 - d. $(\pm 5, 0)$
- 52 For the ellipse $\frac{x^2}{1369} + \frac{y^2}{1225} = 1$ (horizontal major axis), find the coordinates of the foci.
- a. $(\pm 37, 0)$
 - b. $(0, \pm 12)$
 - c. $(\pm 35, 0)$
 - d. $(\pm 12, 0)$
- 53 For the ellipse $\frac{x^2}{1369} + \frac{y^2}{144} = 1$ (horizontal major axis), find the coordinates of the foci.
- a. $(\pm 12, 0)$
 - b. $(\pm 37, 0)$
 - c. $(0, \pm 35)$
 - d. $(\pm 35, 0)$
- 54 A hyperbola has vertices at $(\pm 20, 0)$ and $b = 21$. Find the coordinates of the foci.
- a. $(\pm 29, 0)$
 - b. $(\pm 20, 0)$
 - c. $(\pm 21, 0)$
 - d. $(\pm 41, 0)$

- 55 A hyperbola has vertices at $(\pm 3, 0)$ and $b = 4$. Find the coordinates of the foci.
- a. $(\pm 5, 0)$
 - b. $(\pm 7, 0)$
 - c. $(\pm 3, 0)$
 - d. $(\pm 4, 0)$
- 56 A hyperbola has vertices at $(\pm 9, 0)$ and $b = 40$. Find the coordinates of the foci.
- a. $(\pm 9, 0)$
 - b. $(\pm 41, 0)$
 - c. $(\pm 40, 0)$
 - d. $(\pm 49, 0)$
- 57 A hyperbola has vertices at $(\pm 12, 0)$ and $b = 35$. Find the coordinates of the foci.
- a. $(\pm 12, 0)$
 - b. $(\pm 35, 0)$
 - c. $(\pm 37, 0)$
 - d. $(\pm 47, 0)$
- 58 A hyperbola has vertices at $(\pm 8, 0)$ and $b = 15$. Find the coordinates of the foci.
- a. $(\pm 15, 0)$
 - b. $(\pm 8, 0)$
 - c. $(\pm 23, 0)$
 - d. $(\pm 17, 0)$

- 59 Find the sum of the infinite geometric series with first term $a_1 = 12$ and common ratio $r = \frac{1}{3}$, using $S = \frac{a_1}{1-r}$.
- a. 9
 - b. 36
 - c. 18
 - d. 8
- 60 Find the sum of the infinite geometric series with first term $a_1 = 11$ and common ratio $r = -\frac{1}{4}$, using $S = \frac{a_1}{1-r}$.
- a. $\frac{55}{4}$
 - b. $\frac{44}{5}$
 - c. -44
 - d. $\frac{44}{3}$
- 61 Find the sum of the infinite geometric series with first term $a_1 = 20$ and common ratio $r = \frac{3}{4}$, using $S = \frac{a_1}{1-r}$.
- a. 80
 - b. $\frac{80}{7}$
 - c. 5
 - d. $\frac{80}{3}$
- 62 Find the sum of the infinite geometric series with first term $a_1 = 12$ and common ratio $r = \frac{1}{4}$, using $S = \frac{a_1}{1-r}$.
- a. 9
 - b. 48
 - c. $\frac{48}{5}$
 - d. 16

- 63 Find the sum of the infinite geometric series with first term $a_1 = 15$ and common ratio $r = \frac{3}{4}$, using $S = \frac{a_1}{1-r}$.
- a. $\frac{15}{4}$
 - b. 20
 - c. 60
 - d. $\frac{60}{7}$